



Semester 2 Examinations 2025-2026

Course Instance 4BCT1, 4BS2
Code(s)
Exam(s) Fourth B.Sc. Computer Science and IT
Fourth Science
Module Code(s) CT421
Module(s) Artificial Intelligence

Paper No. 1

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Instructions: Answer any 3 questions. All questions are equally weighted.

Duration 2 hours
No. of Pages 4
Discipline(s) Computer Science
Course Co-ordinator(s) Dr. C. O’Riordan

Requirements:

Release in Exam Venue	No []	Yes [<input]]<="" checked="" td="" type="checkbox"/>
MCQ Answer sheet	No [☒]	Yes []
Handout	No [☒]	Yes []
Formulae & Tables*	No [☒]	Yes []
Cambridge Tables 2 nd Edition**	No [☒]	Yes []
Graph Paper*** A4 Graph Paper 1mm 0.1cm Squared (Standard)	No [☒]	Yes []
Other Materials	No [☒]	Yes []
Graphic material in colour	No [☒]	Yes []

End of requirements.

PTO

Q.1.

- (a) Explain the *iterative deepening* algorithm. Discuss the performance of this algorithm in terms of:

Time complexity
Space complexity
Completeness

(6 marks)

- (b) Two players (MAX and MIN) take turns removing stones from a pile of stones. There are initially **4 stones**. On each turn a player may remove **1 or 2 stones**. The player who takes the **last stone wins**. Player MAX moves first.

- i. Draw the complete game tree for the game, labelling each terminal node as W (win for MAX) or L (loss for MAX). (4 marks)
- ii. Apply minimax ($W=1$, $L=0$) and show the propagated values at every internal node. (3 marks)
- iii. Using your tree from (i), show which nodes can be pruned using alpha-beta pruning. Justify your answer. (4 marks)

- (c) The NK model is a parameterised fitness landscape generator widely used in evolutionary computation research. Explain the roles of the parameters N and K in the NK model, describe how a fitness value is computed for a given binary string of length N, and discuss how varying K affects the ruggedness of the resulting landscape. (8 marks)

PTO

Q.2.

- (a) Explain the concept of diversity in the context of genetic algorithms. Why is maintaining diversity important in an evolutionary search? Discuss how selection pressure, crossover, and mutation each influence population diversity, and describe two mechanisms that can be used to explicitly preserve diversity during a search.

(11 marks)

- (b) Given an undirected graph G with N nodes and E edges, we wish to assign a colour to each node such that no two adjacent nodes share the same colour and the total number of colours used is minimised.

Explain how you would apply a genetic algorithm to find a solution. Your answer should include a description of a suitable chromosome representation, fitness function that penalises adjacent nodes sharing a colour and rewards solutions using fewer colours, and a description of suitable crossover and mutation operators. Discuss how your approach handles the trade-off between minimising colour count and satisfying the colouring constraint.

(14 marks)

Q.3.

- (a) Using an auction setting as your context, describe how agents could interact through structured message passing, giving examples of the types of messages exchanged. Outline any limitations of your approach.

(6 marks)

- (b) Explain the concept of a Nash equilibrium in the context of a two-player game. Using a payoff matrix of your own construction, demonstrate a game that has a single Nash equilibrium, and separately demonstrate a game that has multiple Nash equilibria. Discuss what the existence of multiple equilibria implies for rational agents trying to coordinate their behaviour.

(8 marks)

- (c) Explain what is meant by the prisoner's dilemma and why it presents a challenge for the emergence of cooperation among rational agents. Discuss any two mechanisms that have been proposed to promote cooperation.

(11 marks)

PTO

Q.4.

- (a) Ant Colony Optimisation (ACO) is often described as an artificial life inspired algorithm. Show how ACO, could be used to tackle the Travelling Salesperson Problem. What properties of artificial life are present in the ACO approach. (9 marks)
- (b) Explain briefly the importance of *explainability* in artificial intelligence. With respect to deep learning applied to image classification, outline approaches that can be used to provide explanations. (7 marks)
- (c) LLMs are known to hallucinate — that is, to generate plausible-sounding but factually incorrect or entirely fabricated information. Explain how hallucination occurs, describe the conditions under which it is most likely to arise, and discuss two approaches that have been proposed to detect or mitigate it. (9 marks)

The End